THE CATEGORY OF PRESENTATIONS OF A SCHEME OVER A RING IS FIBRED OVER THE ZARISKI SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement refers to the situation of the preceding construction; the proof gives the explicit base change and asserts its cartesian property, which is a direct verification in this category.

Editorial note. The counted spans carry no external references, so no navigation stubs are needed beyond the ones implied by the statement wording.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The construction of the category C_{X/Lambda} and its objects. used at proof lines 4251-4253. Fixes the category whose fibred structure is asserted.
- The definition of a fibred category and of strongly cartesian morphisms. used at proof lines 4259-4261. Is the property verified by the explicit base change.
- Composition of the structure maps in the category. used at proof lines 4256-4259. Produces the object over U'.

Lemma 0.2. *In the situation above the category $\mathcal{C}_{X/\Lambda}$ is fibred over X_{Zar}.*

Proof. Given an object $U \to \mathbf{A}$ of $\mathcal{C}_{X/\Lambda}$ and a morphism $U' \to U$ of X_{Zar} consider the object $U' \to \mathbf{A}$ of $\mathcal{C}_{X/\Lambda}$ where $U' \to \mathbf{A}$ is the composition of $U \to \mathbf{A}$ and $U' \to U$. The morphism $(U' \to \mathbf{A}) \to (U \to \mathbf{A})$ of $\mathcal{C}_{X/\Lambda}$ is strongly cartesian over X_{Zar}. □

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